

Leveraging Hadoop, Mapreduce, Apache Hive and PySpark for Banking Analytics on GCP Managed Apache Spark(Dataproc)

Project Overview: Analyzing Banking Data

This initiative demonstrates the methodology for processing and examining extensive banking datasets. By leveraging robust frameworks such as Hadoop and MapReduce via Python's `mrjob`, alongside Apache Hive, operations are conducted within GCP Managed Apache Spark (formerly Dataproc).

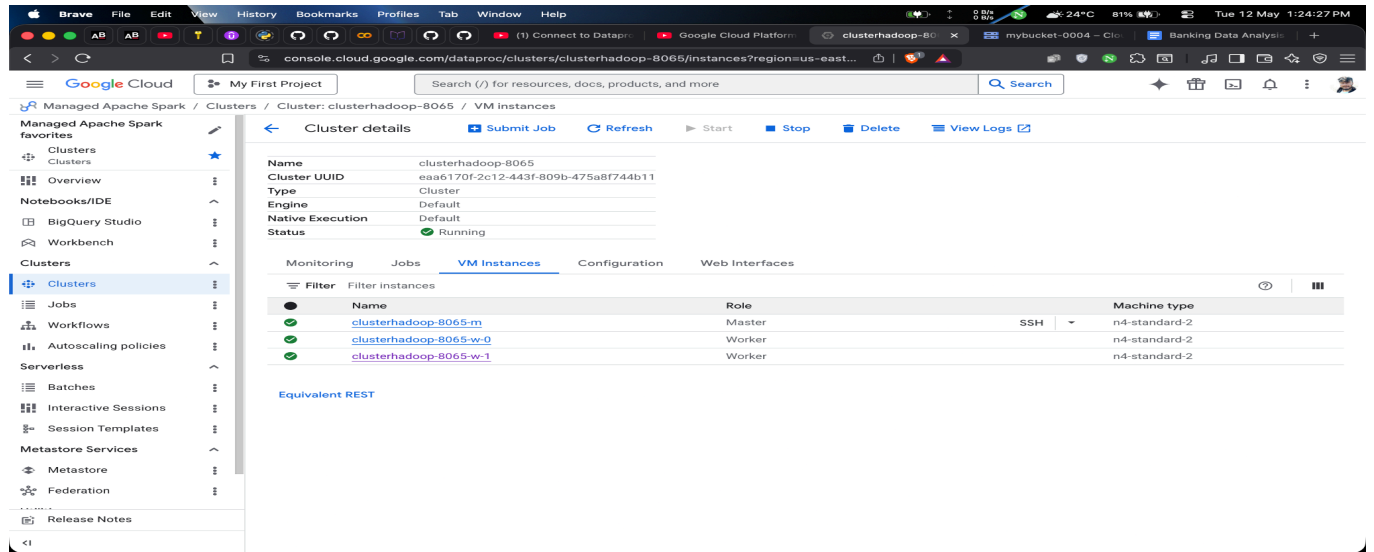
Core Project Deliverables:

- **File Management** → Utilizing the Hadoop Distributed File System (HDFS)
- **MapReduce Execution** → Processing workloads using Python-based `mrjob`
- **Data Warehousing** → SQL-driven analytics through Apache Hive
- **Cloud Orchestration** → Using PySpark for enhanced GCP operations

Step 1: GCP Managed Apache Spark (formerly Dataproc) Cluster Creation

Accessing Managed Apache Spark (formerly Dataproc)

- **GCP Console Navigation** → Select [Managed Apache Spark](#) and then Clusters
- **Initiate Setup** → Click the Create Cluster button
- **Configuration** → Apply the specific parameters listed below:



Cluster description :

The cluster description specifies the exact configuration for the Managed Apache Spark environment on Google Cloud. This includes setting the region, selecting the appropriate image version, and defining the machine types and count for the master and worker nodes. This setup ensures the environment is provisioned with the necessary computational resources and tools, such as JUPYTER and HIVE_WEBHCAT, required for executing the banking analytics workloads.

Parameter	Value
Cluster Name	clusterhadoop-8065
Region	us-east1
Zone	us-east1-d
Image Version	2.3.29-debian12
Master Node Type	n4-standard-2 Standard (1 master, 2 workers)
Worker Nodes	2
Worker Type	n4-standard-2
Optional Components	JUPYTER ,HIVE_WEBHCAT, ICEBERG

Additional settings :

The screenshot shows the Google Cloud console interface for a Managed Apache Spark cluster. The left sidebar contains a navigation menu with options like Clusters, Overview, Notebooks/IDE, BigQuery Studio, Workbench, Clusters (selected), Jobs, Workflows, Autoscaling policies, Serverless, Batches, Interactive Sessions, Session Templates, Metastore Services, Metastore, Federation, and Release Notes. The main panel displays the 'Cluster details' for 'clusterhadoop-8065' in the 'us-east1' region. The configuration table lists various settings for the master and worker nodes.

Setting	Value
Region	us-east1
Zone	us-east1-d
Image version	2.3.29-debian12
Autoscaling	Off
Dataproc Metastore	None
Scheduled deletion	Off
Scheduled stop	Off
Confidential computing enabled?	Disabled
Master node	Standard (1 master, N workers)
Machine type	n4-standard-2
Number of GPUs	0
Primary disk type	hyperdisk-balanced
Primary disk size	100GB
Local SSDs	0
Worker nodes	2
Machine type	n4-standard-2
Number of GPUs	0
Primary disk type	hyperdisk-balanced
Primary disk size	200GB
Local SSDs	0
Secondary worker nodes	0
Secure Boot	Disabled
VTM	Disabled
Integrity Monitoring	Disabled
Cloud Storage staging bucket	mybucket-0004
Network	default
Network tags	None
Internal IP only	No
Project access	Allow API access to all Google Cloud services in the same project

Step 2: Upload Dataset to HDFS

- The primary goal is to prepare the necessary data for processing.
- The core action is to upload the banking dataset (bank.csv) to the Hadoop Distributed File System (HDFS).
- Connect to the master node via SSH to execute specific shell commands.
- Verify the cluster information and confirm the successful loading of the dataset into the HDFS environment.

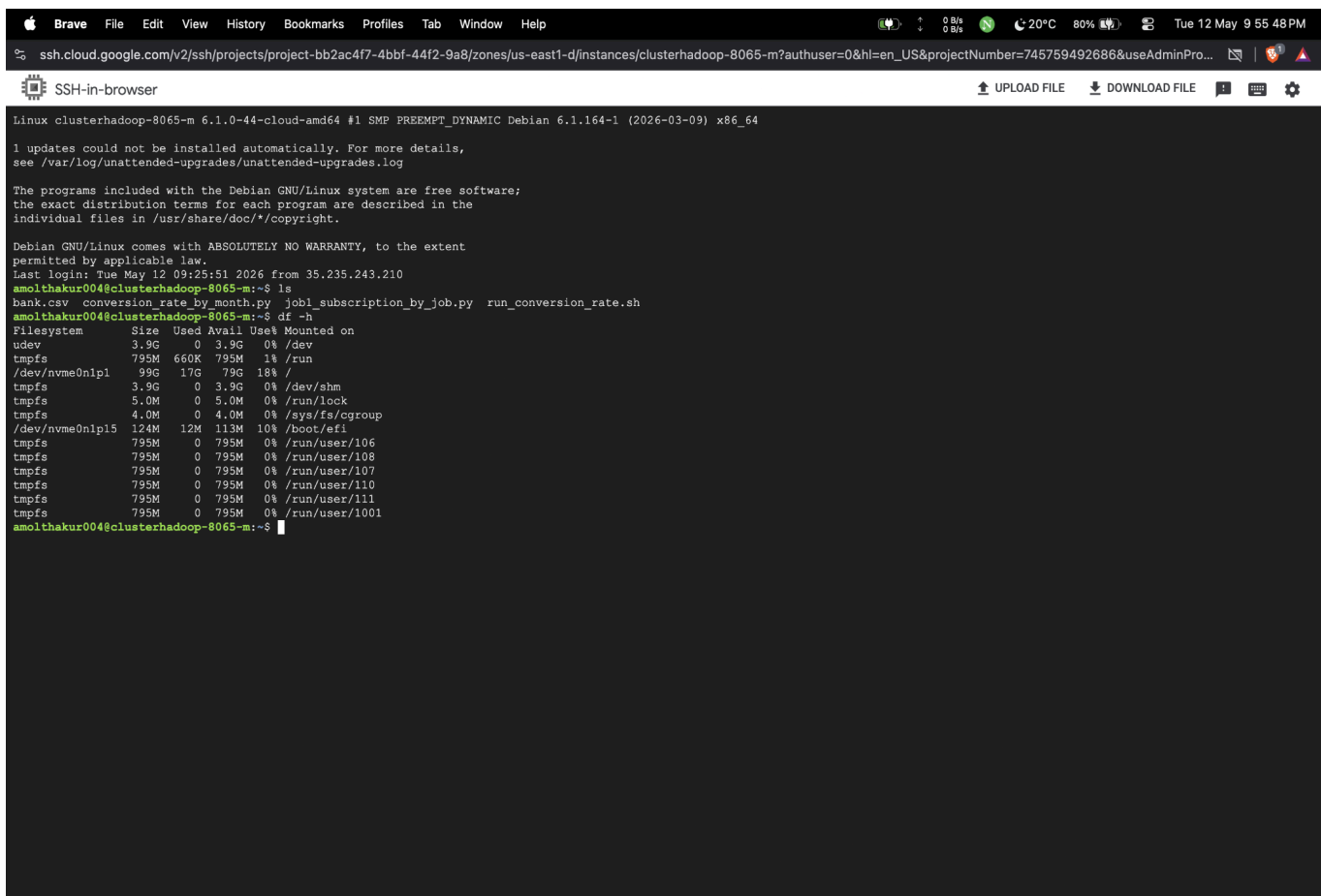
Commands to check the loaded dataset (bank.csv) and cluster information using SSH and to connect with master node :

Shell

```
$ ls
```

Shell

```
$ df -h
```



```
Brave File Edit View History Bookmarks Profiles Tab Window Help
ssh.cloud.google.com/v2/ssh/projects/project-bb2ac4f7-4bbf-44f2-9a8/zones/us-east1-d/instances/clusterhadoop-8065-m?authuser=0&hl=en_US&projectNumber=745759492686&useAdminPro...
SSH-in-browser
UPLOAD FILE DOWNLOAD FILE

Linux clusterhadoop-8065-m 6.1.0-44-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.164-1 (2026-03-09) x86_64

1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log

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permitted by applicable law.
Last login: Tue May 12 09:25:51 2026 from 35.235.243.210
amolthakur004@clusterhadoop-8065-m:~$ ls
bank.csv  conversion_rate_by_month.py  job1_subscription_by_job.py  run_conversion_rate.sh
amolthakur004@clusterhadoop-8065-m:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            3.9G   0  3.9G   0% /dev
tmpfs           795M  660K  795M   1% /run
/dev/nvme0n1p1  99G   17G   79G  18% /
tmpfs           3.9G   0  3.9G   0% /dev/shm
tmpfs           5.0M   0  5.0M   0% /run/lock
tmpfs           4.0M   0  4.0M   0% /sys/fs/cgroup
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tmpfs           795M   0  795M   0% /run/user/111
tmpfs           795M   0  795M   0% /run/user/1001
amolthakur004@clusterhadoop-8065-m:~$
```

Step 3: Run MapReduce Jobs using mrjob

This process involves executing four distinct data analysis workloads using MapReduce via the **mrjob** framework. These jobs are designed to extract key insights from the banking dataset by aggregating data across different criteria. MapReduce is a programming model that divides a large task into two primary phases:

- **Map Phase (Mapper):** This function takes the input data and processes it into key-value pairs. For example, for the average balance jobs, the Mapper extracts the grouping key (like "marital status") and the corresponding value ("balance").
- **Reduce Phase (Reducer):** This function groups the key-value pairs from the Map Phase and performs aggregation. For example, the Reducer groups all records by marital status and calculates the average balance for that group.

The specific jobs executed in this step analyze the total subscriptions grouped by job, the monthly conversion rate, and the average user balances based on marital status and education.

job1_subscription_by_job.py

Python

```
from mrjob.job import MRJob
import csv

class Job1SubscriptionByJob(MRJob):
    def mapper(self, _, line):
        try:
            row = next(csv.reader([line]))
            if row[0] == 'age': # header check
                return
            job = row[1]
            y = row[-1].strip().lower()
            if y == 'yes':
                yield job, 1
        except Exception as e:
            pass # ignore badly formatted lines

    def reducer(self, job, counts):
        yield job, sum(counts)

if __name__ == '__main__':
    Job1SubscriptionByJob.run()
```

conversion_rate_by_month.py

Python

```
from mrjob.job import MRJob
import csv

class Job3ConversionRateByMonth(MRJob):

    def mapper(self, _, line):
        try:
            row = next(csv.reader([line]))
            if row[0] == 'age':
                return
            month = row[10].lower()
            y = row[-1].strip().lower()
            is_yes = 1 if y == 'yes' else 0
            yield month, (1, is_yes)
        except Exception:
            pass

    def reducer(self, month, values):
        total_contacts = 0
        total_yes = 0
        for cnt, is_conversion in values:
            total_contacts += cnt
            total_yes += is_conversion
        if total_contacts > 0:
            conversion_rate = (total_yes / total_contacts) * 100
            yield month, f"{conversion_rate:.2f}%"

if __name__ == '__main__':
    Job3ConversionRateByMonth.run()
```

Avg_balance_marital.py

Python

```
from mrjob.job import MRJob
import csv

class Job2AvgBalanceMarital(MRJob):

    def mapper(self, _, line):
        try:
            row = next(csv.reader([line]))
            # Skip header
            if row[0] == 'age':
                return

            marital_status = row[2]
            balance = float(row[5])

            # Yield marital status as key and balance as value
            yield marital_status, balance
        except Exception:
            pass

    def reducer(self, marital_status, balances):
        total_balance = 0
        count = 0
        for b in balances:
            total_balance += b
            count += 1

        # Yield the final average for the marital status
        yield marital_status, round(total_balance / count, 2)

if __name__ == '__main__':
    Job2AvgBalanceMarital.run()
```

avg_balance_by_education.py

Python

```
from mrjob.job import MRJob
import csv

class Job2AvgBalanceByEducation(MRJob):

    def mapper(self, _, line):
        try:
            row = next(csv.reader([line]))
            if row[0] == 'age': # Skip header
                return
            education = row[3]
            balance = int(row[5])
            yield education, (balance, 1)
        except Exception:
            pass # skip any bad rows

    def reducer(self, education, values):
        total_balance = 0
        count = 0
        for bal, cnt in values:
            total_balance += bal
            count += cnt
        if count > 0:
            avg = total_balance / count
            yield education, round(avg, 2)

if __name__ == '__main__':
    Job2AvgBalanceByEducation.run()
```


MapReduce Jobs Analysis

Job description		Python file with path in bucket	Output path in bucket	Command to view results
Total number of subscriptions by job		mybucket-0004/Jobs / job1_subscription_by_job.py	/gs://mybucket-0004/output_results_1778480600_v3	\$ gsutil cat gs://mybucket-0004/output_results_1778480600_v3/part-*
Conversion rate by every month		mybucket-0004/Jobs / conversion_rate_by_month.py	gs://mybucket-0004/output_conversion_rate/part-0000	\$ gsutil cat gs://mybucket-0004/output_conversion_rate/part-*
Average balance of users based on their marital status		mybucket-0004/Jobs / Avg_balance_marital.py	gs://mybucket-0004/output_avg_balance/part-00000	\$ gsutil cat gs://mybucket-0004/output_avg_balance/part-*
Average balance based on education of user		mybucket-0004/Jobs / avg_balance_by_education.py	gs://mybucket-0004/output_avg_education/part-0000	\$ gsutil cat gs://mybucket-0004/output_avg_education/part-*

Output on SSH:

Shell

```
$ gsutil cat gs://mybucket-0004/output_results_1778480600_v3/part-*
```

```
Brave File Edit View History Bookmarks Profiles Tab Window Help 1 KB/s 14 KB/s 20°C 80% Tue 12 May 11 49 32 PM
ssh.cloud.google.com/v2/ssh/projects/project-bb2ac4f7-4bbf-44f2-9a8/zones/us-east1-d/instances/clusterhadoop-8065-m?authuser=0&hl=en_US&projectNumber=745759492686&useAdminPro...
SSH-in-browser UPLOAD FILE DOWNLOAD FILE
Linux clusterhadoop-8065-m 6.1.0-44-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.164-1 (2026-03-09) x86_64

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bank.csv conversion_rate_by_month.py job1_subscription_by_job.py run_conversion_rate.sh
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tmpfs           5.0M   0 5.0M   0% /run/lock
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"blue-collar" 69
"services" 38
"student" 19
"admin." 58
"self-employed" 20
"technician" 83
"unemployed" 13
"entrepreneur" 15
"housemaid" 14
"management" 131
"retired" 54
"unknown" 7
amolthakur004@clusterhadoop-8065-m:~$
```

Shell

```
$ gsutil cat gs://mybucket-0004/output_conversion_rate/part-*
```

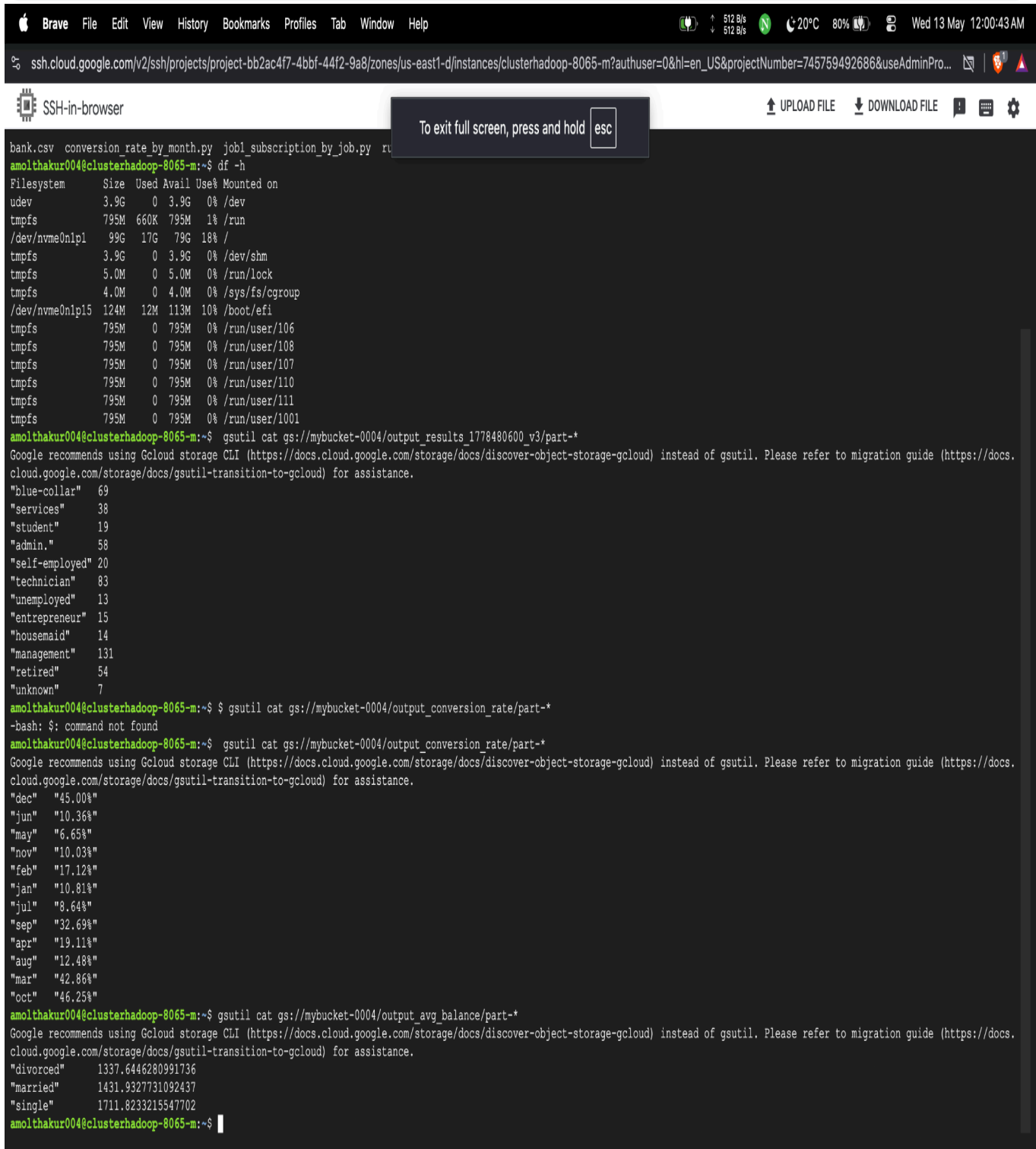
```
Brave File Edit View History Bookmarks Profiles Tab Window Help 512 B/s 1 KB/s 20°C 80% Tue 12 May 11 54 16 PM
ssh.cloud.google.com/v2/ssh/projects/project-bb2ac4f7-4bbf-44f2-9a8/zones/us-east1-d/instances/clusterhadoop-8065-m?authuser=0&hl=en_US&projectNumber=745759492686&useAdminPro...
SSH-in-browser UPLOAD FILE DOWNLOAD FILE ! ...

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tmpfs           3.9G   0 3.9G   0% /dev/shm
tmpfs           5.0M   0 5.0M   0% /run/lock
tmpfs           4.0M   0 4.0M   0% /sys/fs/cgroup
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tmpfs           795M   0 795M   0% /run/user/106
tmpfs           795M   0 795M   0% /run/user/108
tmpfs           795M   0 795M   0% /run/user/107
tmpfs           795M   0 795M   0% /run/user/110
tmpfs           795M   0 795M   0% /run/user/111
tmpfs           795M   0 795M   0% /run/user/1001
amolthakur004@clusterhadoop-8065-m:~$ gsutil cat gs://mybucket-0004/output_results_1778480600_v3/part-*
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"technician" 83
"unemployed" 13
"entrepreneur" 15
"housemaid" 14
"management" 131
"retired" 54
"unknown" 7
amolthakur004@clusterhadoop-8065-m:~$ gsutil cat gs://mybucket-0004/output_conversion_rate/part-*
-bash: gs: command not found
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"dec" "45.00%"
"jun" "10.36%"
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"nov" "10.03%"
"feb" "17.12%"
"jan" "10.81%"
"jul" "8.64%"
"sep" "32.69%"
"apr" "19.11%"
"aug" "12.48%"
"mar" "42.86%"
"oct" "46.25%"
amolthakur004@clusterhadoop-8065-m:~$
```

Shell

```
$ gsutil cat gs://mybucket-0004/output_avg_balance/part-*
```



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ssh.cloud.google.com/v2/ssh/projects/project-bb2ac4f7-4bbf-44f2-9a8/zones/us-east1-d/instances/clusterhadoop-8065-m?authuser=0&hl=en_US&projectNumber=745759492686&useAdminPro...

SSH-in-browser

To exit full screen, press and hold **esc**

bank.csv conversion_rate_by_month.py job1_subscription_by_job.py ru

amolthakur004@clusterhadoop-8065-m:~\$ df -h

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	3.9G	0	3.9G	0%	/dev
tmpfs	795M	660K	795M	1%	/run
/dev/nvme0n1p1	99G	17G	79G	18%	/
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tmpfs	5.0M	0	5.0M	0%	/run/lock
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amolthakur004@clusterhadoop-8065-m:~\$ gsutil cat gs://mybucket-0004/output_results_1778480600_v3/part-*

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```
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amolthakur004@clusterhadoop-8065-m:~\$ gsutil cat gs://mybucket-0004/output_conversion_rate/part-*

-bash: \$: command not found

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```

amolthakur004@clusterhadoop-8065-m:~\$ gsutil cat gs://mybucket-0004/output_avg_balance/part-*

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```
"divorced" 1337.6446280991736
"married" 1431.9327731092437
"single" 1711.8233215547702
```

amolthakur004@clusterhadoop-8065-m:~\$

Shell

```
$ gsutil cat gs://mybucket-0004/output_avg_education/part-*
```

```
Brave File Edit View History Bookmarks Profiles Tab Window Help
512 B/s 512 B/s 20°C 80% Wed 13 May 12 03 50 AM
ssh.cloud.google.com/v2/ssh/projects/project-bb2ac4f7-4bbf-44f2-9a8/zones/us-east1-d/instances/clusterhadoop-8065-m?authuser=0&hl=en_US&projectNumber=745759492686&useAdminPro...
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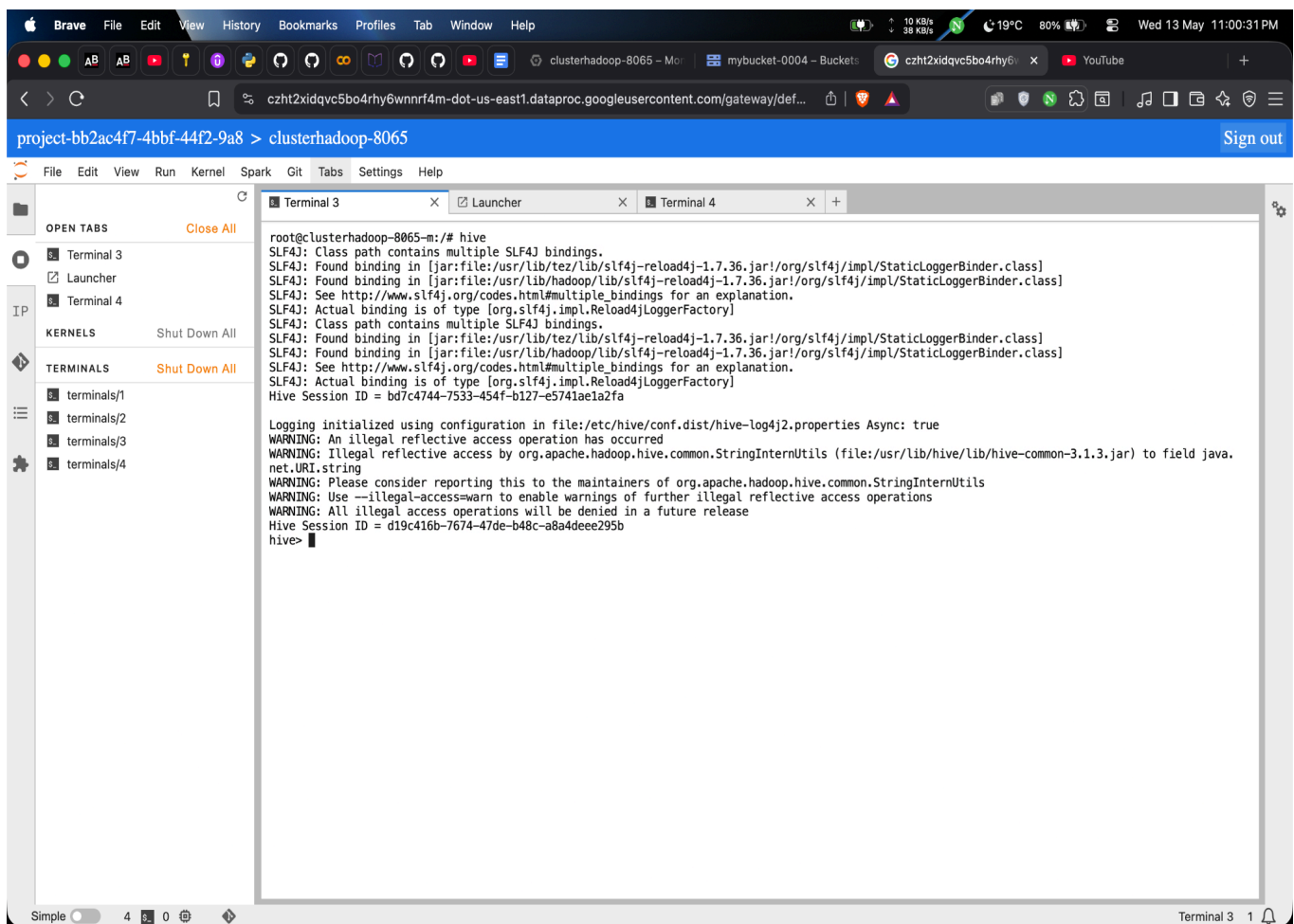
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amolthakur004@clusterhadoop-8065-m:~$ $ gsutil cat gs://mybucket-0004/output_conversion_rate/part-*
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"unknown" 1701.2459893048128
"primary" 1411.5442477876106
"secondary" 1196.8143972246314
"tertiary" 1775.4237037037037
amolthakur004@clusterhadoop-8065-m:~$
```

Step 4: Setting up Apache Hive and Queries

Apache Hive is a data warehouse software built on top of Apache Hadoop for providing data query and analysis. It allows users to read, write, and manage large datasets residing in distributed storage using a SQL-like interface called HiveQL. Hive simplifies big data analysis by automatically converting these queries into executable jobs, such as MapReduce or Spark, making it highly accessible for those familiar with standard SQL.

Launch Hive

```
Shell  
$ hive
```



The screenshot shows a web browser window with a terminal interface. The terminal displays the output of the 'hive' command. The logs show SLF4J warnings about multiple bindings and a Hive Session ID. The terminal output is as follows:

```
root@clusterhadoop-8065-m:/# hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/tez/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/tez/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
Hive Session ID = bd7c4744-7533-454f-b127-e5741ae1a2fa

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.hive.common.StringInternUtils (file:/usr/lib/hive/lib/hive-common-3.1.3.jar) to field java.net.URI.string
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.hive.common.StringInternUtils
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
Hive Session ID = d19c416b-7674-47de-b48c-a8a4deee295b
hive>
```

Create Table

SQL

```
CREATE EXTERNAL TABLE IF NOT EXISTS bank_data (  
    age INT,  
    job STRING,  
    marital STRING,  
    education STRING,  
    is_default STRING,  
    balance INT,  
    housing STRING,  
    loan STRING,  
    contact STRING,  
    day INT,  
    month STRING,  
    duration INT,  
    campaign INT,  
    pdays INT,  
    previous INT,  
    poutcome STRING,  
    y STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE  
LOCATION 'gs://mybucket-0004/bankdata/'  
TBLPROPERTIES ("skip.header.line.count"="1");
```

This process establishes an external table named `bank_data` in Hive. It defines a schema for the banking dataset (including columns like `age`, `job`, and `balance`) and links it directly to comma-separated text files located in a specified Google Cloud Storage path (`gs://mybucket-0004/bankdata/`). Because it is an `EXTERNAL` table, the underlying source data remains intact even if the Hive table definition is deleted.

Hive Queries

Query 1: Average balance grouped by job

SQL

```
SELECT job, AVG(balance) FROM bank_data GROUP BY job;
```

project-bb2ac4f7-4bbf-44f2-9a8 > clusterhadoop-8065

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OPEN TABS Close All

- Terminal 3
- Terminal 5
- Terminal 4

KERNELS Shut Down All

TERMINALS Shut Down All

- terminals/1
- terminals/2
- terminals/3
- terminals/4
- terminals/5

Terminal 3

```
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
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WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.hive.common.StringInternUtils
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
Hive Session ID = d19c416b-7674-47de-b48c-a8a4deee295b
hive> SELECT job, AVG(balance) FROM bank_data GROUP BY job;
Query ID = root_20260513185341_6d6dd81a-f9dd-47fc-8785-c6bca17a4f95
Total jobs = 1
Launching Job 1 out of 1
Tez session was closed. Reopening...
Session re-established.
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1778598041902_0029)
```

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0

VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 9.38 s

OK

```
admin. 1226.73640167364
blue-collar 1085.161733615222
entrepreneur 1645.125
housemaid 2083.8035714285716
management 1766.9287925696594
retired 2319.191304347826
self-employed 1392.4098360655737
services 1103.9568345323742
student 1543.8214285714287
technician 1330.99609375
unemployed 1089.421875
unknown 1501.7105263157894
Time taken: 21.37 seconds, Fetched: 12 row(s)
hive>
```

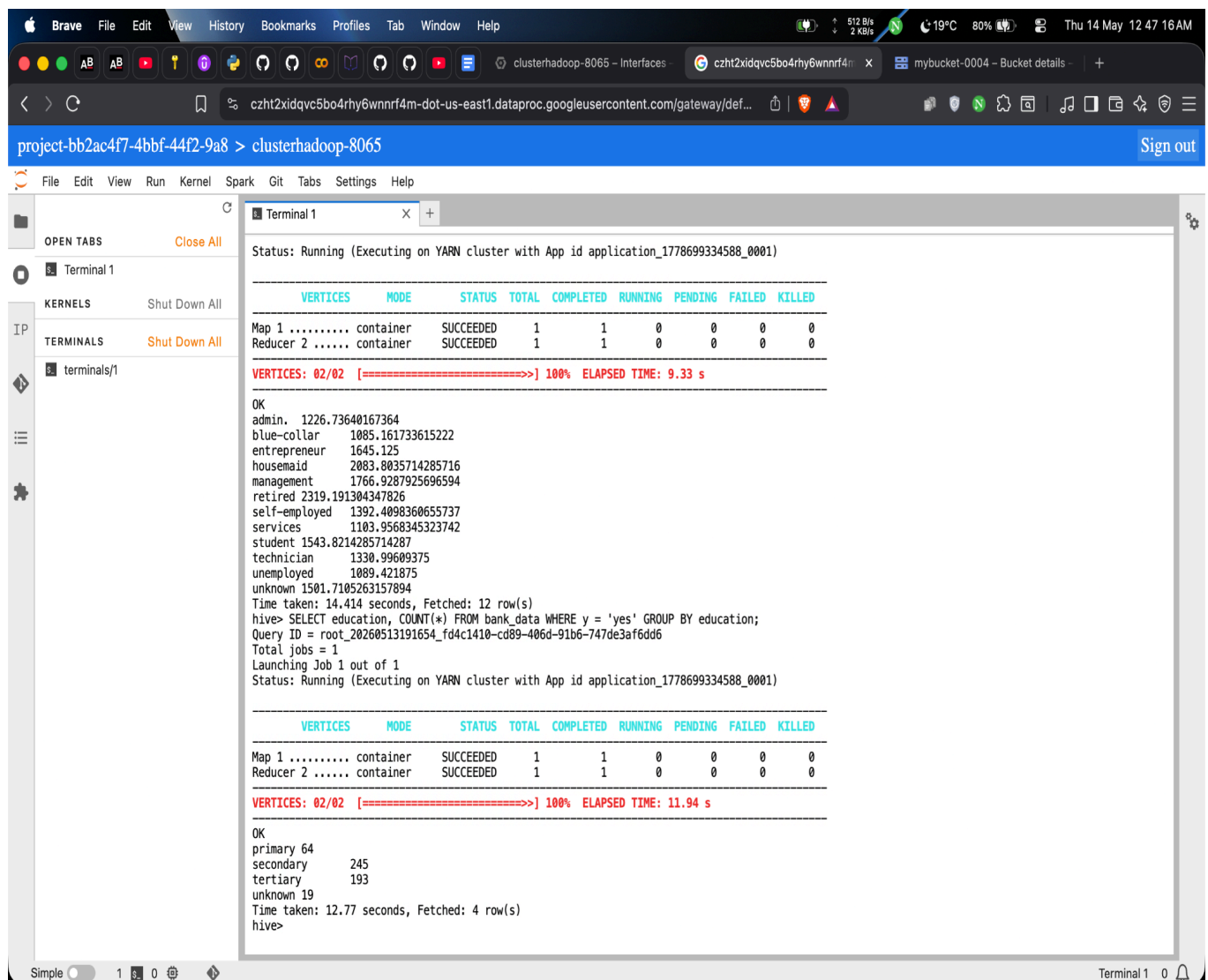
Simple 5 0

Terminal 3 1

Query 2: Count of term deposit subscriptions by education

SQL

```
SELECT education, COUNT(*) FROM bank_data WHERE y = 'yes' GROUP BY education;
```



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Terminal 1

KERNELS [Shut Down All](#)

TERMINALS [Shut Down All](#)

terminals/1

Terminal 1

Status: Running (Executing on YARN cluster with App id application_1778699334588_0001)

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0

VERTICES: 02/02 [=====] 100% ELAPSED TIME: 9.33 s

OK

admin. 1226.73640167364

blue-collar 1085.161733615222

entrepreneur 1645.125

housemaid 2083.8035714285716

management 1766.9287925696594

retired 2319.191304347826

self-employed 1392.4098360655737

services 1103.9568345323742

student 1543.8214285714287

technician 1330.99609375

unemployed 1089.421875

unknown 1501.7105263157894

Time taken: 14.414 seconds, Fetched: 12 row(s)

hive> SELECT education, COUNT(*) FROM bank_data WHERE y = 'yes' GROUP BY education;

Query ID = root_20260513191654_fd4c1410-cd89-406d-91b6-747de3af6dd6

Total jobs = 1

Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1778699334588_0001)

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0

VERTICES: 02/02 [=====] 100% ELAPSED TIME: 11.94 s

OK

primary 64

secondary 245

tertiary 193

unknown 19

Time taken: 12.77 seconds, Fetched: 4 row(s)

hive>

Simple 1 0

Terminal 1 0

Query 3: Month-wise campaign success

SQL

```
SELECT month, COUNT(*) FROM bank_data WHERE y = 'yes' GROUP BY month;
```

The screenshot shows a web browser window with a terminal interface. The terminal displays the execution of a Hive query to count successful campaigns by month. The query is: `SELECT month, COUNT(*) FROM bank_data WHERE y = 'yes' GROUP BY month;`

The terminal output shows the query execution progress and results. It includes a table with columns: VERTICES, MODE, STATUS, TOTAL, COMPLETED, RUNNING, PENDING, FAILED, and KILLED. The results show that the query completed successfully, with 12 rows fetched.

The results are as follows:

Month	Count
apr	56
aug	79
dec	9
feb	38
jan	16
jul	61
jun	55
mar	21
may	93
nov	39
oct	37
sep	17

